

Module 3 Contribution Journal

ITAI 1371 – Machine Learning

Neural Crew

Patrick Ray

For the Module 3 lab, my main contribution was completing my part of the Machine Learning Workflow and Types of Learning notebook. I worked through the notebook in Google Colab and ran the cells so I could see the results as I went through the assignment thoroughly looking for mistakes. Since I am still pretty new to machine learning and Python, I tried not to just run everything without looking at it. I spent time looking at the outputs and trying to understand the layout and what the code was actually doing.

One part I worked on was identifying different types of machine learning. I completed the assessment where I had to decide if different situations were supervised, unsupervised, or reinforcement learning. I got 5 out of 5 corrections on that section. This helped me contribute to the assignment because I was able to show that I understood the differences between the three types instead of just running the code.

I also worked with the Wine dataset and the classification model. One of the parts of the lab allowed me to choose three features and see how well the model performed using them. I selected malic acid, flavonoids, and hues. When I ran the model with those features, I got about 83.3% accuracy. The original model had about 88.9% accuracy. At first, I was a little confused because I thought I was supposed to get a better score, but after looking at the results I understood that changing the features can actually make the model perform better or worse. This was one of the more useful parts of the lab for me because I could actually see how a change I made affected the results.

Another contribution I made was completing the reflection and analysis section inside the notebook. I explained supervised, unsupervised, and reinforcement learning in my own words. I also wrote about why I thought Logistic Regression performed better, what I could try next to improve the model, and how machine learning could be used in a real world situation. For my real world example, I used transportation and self-driving vehicles because I think this is an interesting way machine learning can be used. I wrote about how a vehicle could use things such as camera images, traffic information, road conditions, speeds, and locations to help make driving decisions.

I also made sure the notebook was completed and saved correctly. I ran the assessment and checked that my answers were correct, and I reviewed the reflection section before finishing it. After completing my work, I downloaded the notebook and uploaded my completed Module 3 notebook to my GitHub repository. I made sure the file was named correctly on GitHub so that

my work could be shared with the rest of the group and eventually included with the Neural Crew submission.

Working on my part of the group project also gave me more practice using GitHub. I am still getting used to GitHub, so even uploading the notebook and making sure it was in the correct repository was good practice for me. I feel more comfortable doing it now than I did when we first started using GitHub. I know I still have more to learn with branches, commits, and working with other people's code, but I am starting to understand why GitHub is useful when several people are working on the same project.

Overall, I feel like I contributed by completing and checking my portion of the Module 3 lab, experimenting with the Wine model, completing the assessment and written analysis, and getting my completed notebook onto GitHub for the group. I also tried to understand what I was doing instead of just trying to finish the assignment. There were some parts that I had to slow down on, especially when looking at the code and the model results, but I think that is helping me get more comfortable with machine learning. As we continue with the class, I hope I can contribute more as I get better with Python, GitHub, and building different machine learning models.